(c) G.M. for Continuous grouped data

Let x_i be the mid point of the class interval

$$\text{G. M.} = \text{Antilog}\left[\frac{\sum_{i=l}^{n} f_i \log x_i}{N}\right]$$

Example 5.10

The following is the distribution of marks obtained by 109 students in a subject in an institution. Find the Geometric mean.

Marks	4-8	8-12	12-16	16-20	20-24	24-28	28-32	32-36	36-40
No. of Students	6	10	18	30	15	12	10	6	2

Solution:

Marks	**Mid point (x_i)**	f_i	**log x_i**	$f_i \log x_i$
4-8	6	6	0.7782	4.6692
8-12	10	10	1.0000	10.0000
12-16	14	18	1.1461	20.6298
16-20	18	30	1.2553	37.6590
20-24	22	15	1.3424	20.1360
24-28	26	12	1.4150	16.980
28-32	30	10	1.4771	14.7710
32-36	34	6	1.5315	9.1890
36-40	38	2	1.5798	3.1596
Total		**N =109**		**137.1936**

$$\text{G.M.} = \text{Antilog}\left[\frac{\sum_{i=l}^{n} f_i \log x_i}{N}\right]$$

$$= \text{Antilog}\left[\frac{137.1936}{109}\right] = \text{Antilog}\,[1.2587]$$

$$\text{G. M.} = 18.14$$

Geometric mean marks of 109 students in a subject is 18.14

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